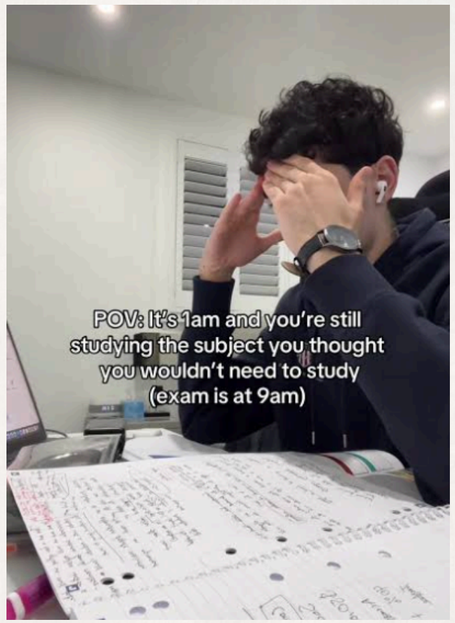
CORTISOL TEST KIT

GROUP A:
VIOLET, MATTHEW, BENJAMIN

CASE 01

BEFORE A PRESENTATION

“What shall I do?”
“Am I in the right state?”



“Shall I sleep?”
“Am I still learning effectively?”

CASE 02

“Did it measure cortisol?”
“Does it ACCURATE?”

CASE 02

“Expensive” “Time-consuming”
“Is there better ALTERNATIVES?”

MARKET OPPORTUNITY

01

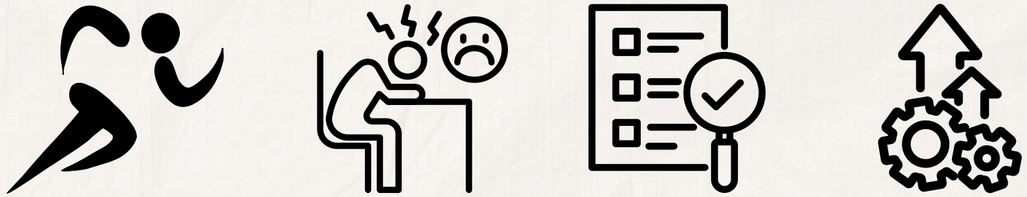
Budget-Conscious Athletes
“monitor physiological trends”
“budget friendly”

02

Health-Conscious Individuals
“body's stress response”

PROBLEM STATEMENT

“Current cortisol testing is costly, time-consuming, and inconvenient, making regular monitoring inaccessible to the general public and budget-conscious athletes.”



OUR SOLUTION

-Non-Invasive
Saliva collection using designated chewing gum.

-Convenient
Quick, light in weight, easy testing anytime, anywhere.

-All-in-One
Compact device for saliva collection and cortisol detection.

-Replaceable Chip
Easy-to-change disposable sensing chip.

-Smart Monitoring
Simple display, LED indicators and mobile app tracking.

-Hygienic
Built-in gum disposal compartment, all parts washable.

-Affordable & Accessible
Low-cost alternative to frequent laboratory testing.

PROGRESS

01.

Product

02.

Market

03.

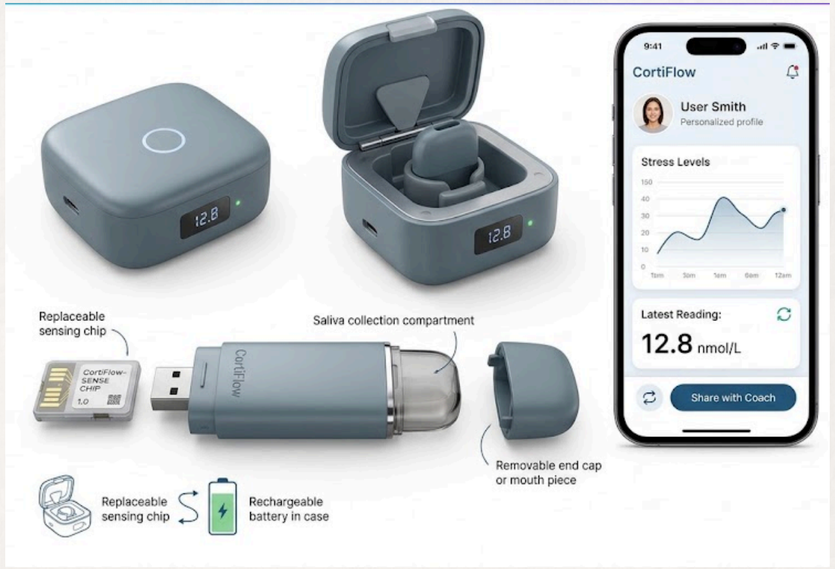
Progress

04.

Regulation

PRODUCT

What is our product’s core competitiveness?
Where does our innovation lie?
what our product should be?



Product

Break the limits of size
Non-invasive
Lightweight and portable
Non-medical grad design

Feature

Extremely low cost per test
High-frequency testing
Lock down source of impact

Market

More comprehensive report (source of impact)
Test report analysis bound consumption patterns
Free test - Paid report

MARKET SIZE

The global cortisol testing kits market was valued approximately US 208 million and is projected to reach US 373 million in 2035

Core Market

Main target consumer groups:

Urban high-stress individuals
- stress tracking

Fitness enthusiasts
- sleep recovery
- muscle repair

Hong Kong

Hong Kong population:	7.5 M
Workplace stress:	60%
Impact of stress on daily life:	84%
Gen Z mental health:	43.5%
Health product retail market:	1.2B

Top-Down Approach

TAM (Total Addressable Market)
HK$ 90–140 million per year

SAM (Serviceable Available Market)
HK$ 140M–210 million per year

SOM (Serviceable Obtainable Market)
HK$ 9.44–21.2 million per year

Summary

Based on the top-down approach, the Total Addressable Market for cortisol testing kits in Hong Kong reaches HK$ 90–140 million per year, with a Serviceable Obtainable Market of approximately HK$ 9.44–21.2 million per year.

IN PROGRESS

The cortisol testing market already exists in Hong Kong, but they have different market strategy and features.

Technology Readiness Level (TRL)
3-4 Laboratory and Product sample proof of concept success.

We already have a preliminary direction for the software and analysis

Current products in the Hong Kong market:
eNano Health、WelSpring、HK BioTek

Priced: HK$200–950
Retail channel: Mannings, Wastons
Even B2C B2B

Where are we going?

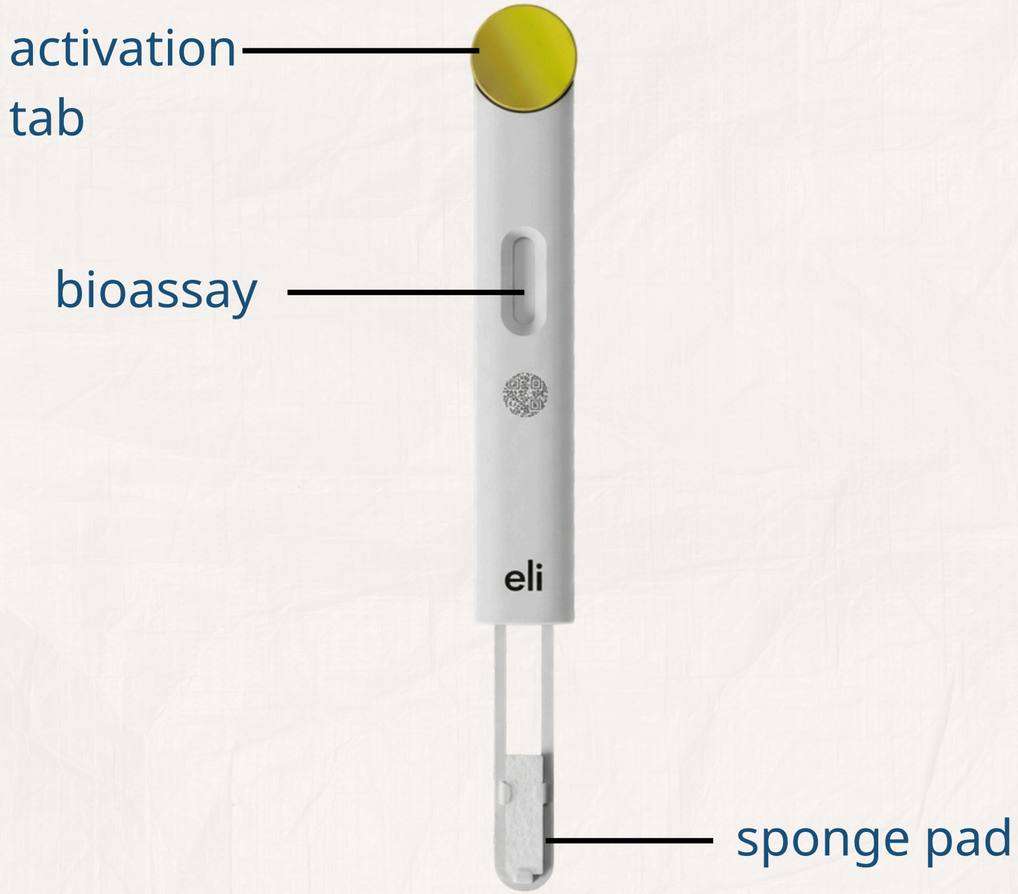
Rapid testing + App analysis Strategy
Make product sketches based on the current design.

Regulation and compliance

Compliance Area	When It Applies	Key Requirements / Risks
Product Classification	All cases	Determines whether the product is a wellness device or an IVD medical device; affects claims, sales channels,
MDACS Medical Device Listing	If claiming cortisol measurement for health assessment, or selling to clinics, hospitals, or	Hong Kong operates a voluntary listing system; government procurement and some healthcare
Advertising & Health Claims	Website, app, packaging, KOLs, social media	Regulated under the Undesirable Medical Advertisements Ordinance and Trade Descriptions
Consumer Product Safety	Hardware, packaging, and user manuals	The Consumer Goods Safety Ordinance requires products to be safe; supplying non-compliant goods is
Electrical Safety	Device contains a rechargeable battery or electronic circuit	Must comply with electrical product safety requirements
Bluetooth / Wireless Communication	Device connects to the app via Bluetooth or Wi-Fi	Must comply with OFCA radio equipment requirements
Personal Data Privacy	App collects cortisol, health, and usage data	Governed by the Personal Data (Privacy) Ordinance; health data is highly sensitive
Data Use Restrictions	Employers, insurers, AI analysis, advertising	Health data must not be used for insurance underwriting, employer evaluation, or advertising
Saliva Sample Hygiene	Sampling components, reagents, consumables	Involves biological sample contact and cross-contamination risks
Post-Market Surveillance	After product launch	Complaints, adverse events, recalls, and ongoing product safety must be managed

COMPETITORS

The wellness sector



EliHealth:

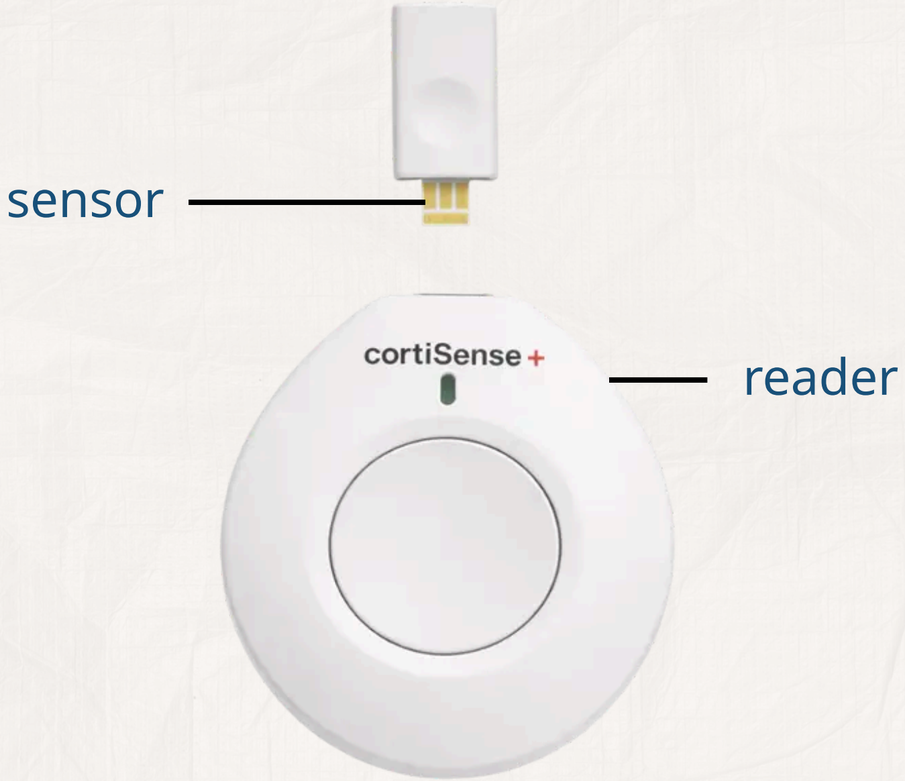
- collects saliva
- ~20 mins results
- upload a photo to app
- suggestions → broad category of struggles

- disposed

→ slow results

COMPETITORS

The wellness sector



Nutrix’s cortiSense+:

- collects saliva
- 3 mins result (claimed)
- sends data to gSense app
- combine glucose levels → health report

- reusable reader
- sensor disposed

→ no specific collection method

comparison

Feature	Eli Health	Nutrix — cortiSense	Our proposed solution
Saliva collection	Hold a test tip in the mouth for 60 seconds	Apply saliva to a sensor (unclear collection steps)	Chew gum, then squeeze out saliva for testing
Result time	**SLOW** ~ 20 minutes	**OKAY** ~3 minutes(claimed)	**FAST** A few minutes → aiming for seconds
Getting results	**SLOW** Scan the test with a phone app	**FAST** no phone scanning	**FAST** no phone scanning
App focus	General wellness, sleep, training and recovery	Broader health monitoring	Focused on training: combines cortisol trends, exercise, rest and stress ratings to support recovery decisions

THANK YOU
VERY MUCH